

Many insects belonging to the order the Orthoptera can be readily identified by their large back legs which enable them to hop or jump.

Locusts are large herbivorous insects that can be serious pests of agriculture due to their ability to form dense and highly mobile swarms. They are species of short-horned grasshoppers that periodically form large populations in dense migrating groups, where individuals differ in several characteristics from those living separately.

In Australia there are three main pest species of locust

- the [Australian plague locust](#) (*Chortoicetes terminifera*)
- the [spur-throated locust](#) (*Austracris guttulosa*)
- the [migratory locust](#) (*Locusta migratoria*)

All three species are native to Australia. Locusts belong to the same order of insects as grasshoppers, katydids and crickets - the Orthoptera.

[expand all]

What is the difference between a locust and a grasshopper?

Locusts and grasshoppers are the same in appearance, but locusts can exist in two different behavioural states (solitary and gregarious), whereas most grasshoppers do not. When the population density is low, locusts behave as individuals, much like grasshoppers. However, when locust population density is high, individuals undergo physiological and behavioural changes, known as phase polyphenism, and they form gregariously behaving bands of nymphs or swarms of adults.

In addition to changes in behaviour, phase change may be accompanied by changes in body shape and colour, and in fertility, physiology and survival. These changes are so dramatic in some species that the swarming and non-swarming forms were once considered to be different species. The scale of population increase and migrations also distinguish those species known as locusts from grasshoppers.

The distinction between locusts and grasshoppers is often not clear-cut, as the extent to which different species exhibit gregarious phase characters is graded. The migratory locust has all of the features associated with phase change - differences in body shape and colour, fertility and gregarious behaviour in both the nymph and adult life stages, forming dense bands and swarms. The Australian plague locust also forms dense nymph bands and adult swarms, but does not exhibit changes in body colour. Spur-

throated locust nymphs do not form bands and the adults do not lay eggs gregariously, but they do form dense swarms.

Some species that are called grasshoppers, such as *Austroicetes cruciata*, *Oedaleus australis* and *Peakesia* spp. can form loose swarms at high densities, but do not generally migrate long distances as locusts do.

Classification of Locusts

Locusts belong to the same order as grasshoppers, katydids and crickets - the Orthoptera (derived from the Greek words orthos meaning straight or rigid and ptera meaning wing).

In Australia there are over 2,800 species of Orthoptera and over 700 species of locusts and grasshoppers. Many species are yet to be scientifically described.

The following table shows how three main pest species of locusts in Australia are currently classified.

Taxonomic category	Australian plague locust	Migratory locust	Spur-throated locust
Kingdom	Animalia	Animalia	Animalia
Phylum	Arthropoda	Arthropoda	Arthropoda
Superclass	Hexapoda	Hexapoda	Hexapoda
Class	Insecta	Insecta	Insecta
Order	Orthoptera	Orthoptera	Orthoptera
Suborder	Caelifera	Caelifera	Caelifera
Superfamily	Acridoidea	Acridoidea	Acridoidea
Family	Acrididae	Acrididae	Acrididae
Subfamily	Oedipodinae	Oedipodinae	Catantopinae
Tribe	-	Locustini	Cyrtacanthacridini*
Genus	Chortoicetes	Locusta	Austracris**
Species	terminifera	migratoria	guttulosa
Subspecies	-	migratorioides	-

* the Cyrtacanthacridini is considered by some taxonomists to be a subfamily of the Acrididae

** *Austracris guttulosa* was formerly classified as *Nomadacris guttulosa*

Further reading

CSIRO, 1991, *The Insects of Australia: a textbook for students and research workers*, Volume 1. 2nd edn, Melbourne, CSIRO & Melbourne University Press.

Pener, MP & Simpson, SJ. 2009, 'Locust phase polyphenism: an update', *Advances in Insect Physiology*, 36, 1-286.

Rentz, DCF, Lewis RC, Su, YN & Upton, MS. 2003, A guide to Australian grasshoppers and locusts, Natural History Publications, (Borneo), Kota Kinabalu.

Growth stages of locust nymphs

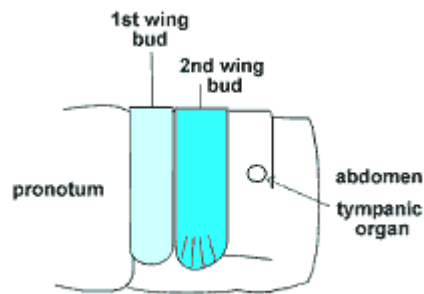
The following diagrams show how to determine the growth stage (instar) of locust or grasshopper nymphs, based on characteristics of the developing wings (adapted from: [Baker, 1984](#) by permission from NSW Department of Primary Industries).

Instar

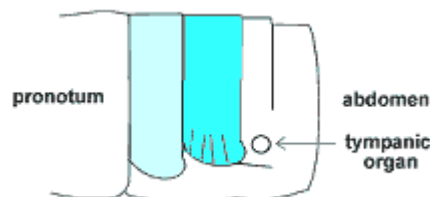
Wing development

Key characteristics

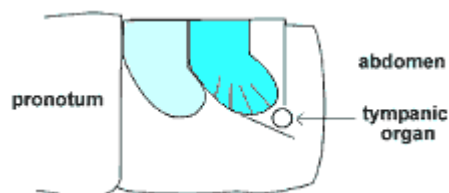
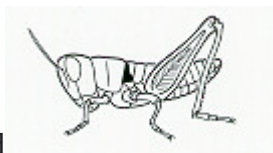
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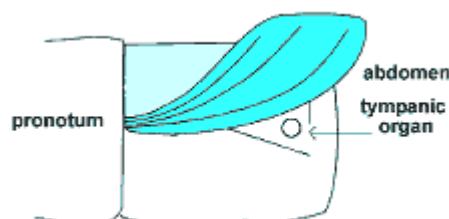
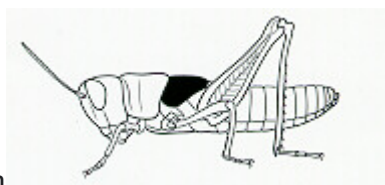
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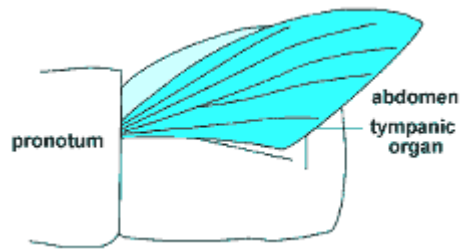
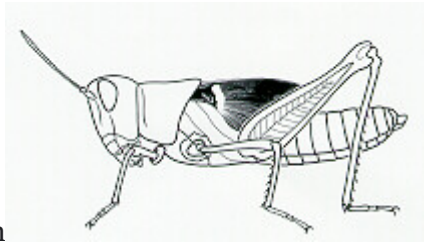
Third



Fourth



Fifth



Note: nymphs of some species of locusts and grasshoppers have more than 5 growth stages (eg the spur-throated locust). However, in these species the wing buds consistently flip upwards in the penultimate instar and the additional instars resemble instars 2 and 3 as shown above.

Identifying male and female locusts

Adult male and female locusts are readily distinguished by the shape of the tip of the abdomen:

Male - tip of abdomen smooth and rounded



Female - tip of abdomen jagged



The end of the abdomen of the male locust is rounded due to the sub-genital plate which conceals the reproductive organs.

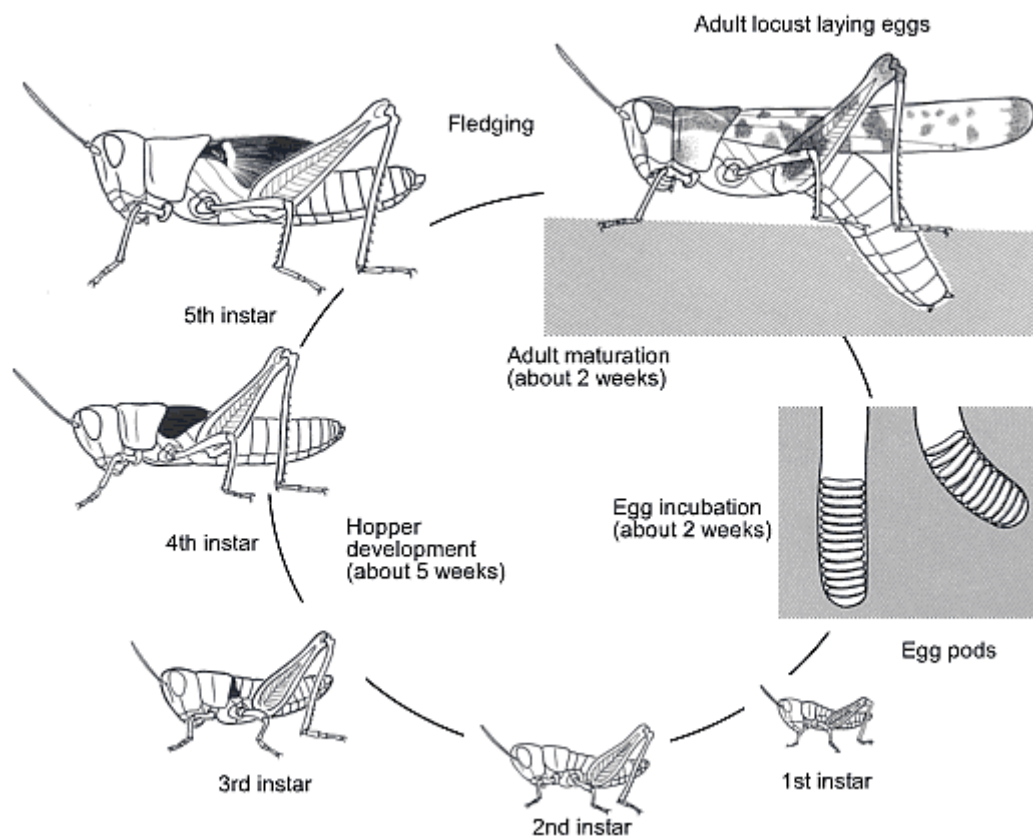
The end of the abdomen of the female locust appears pointed due to the upper and lower jaws of the ovipositor.

On average, adult male locusts are smaller than adult females of the same species. However, size is not a reliable character to determine the sex of a locust as it varies according to the quality and abundance of food during the nymph stage.

Lifecycle of a locust

Locusts undergo incomplete or direct metamorphosis. Unlike in insects such as butterflies or moths, there is no pupal stage and juveniles are similar in appearance to adults.

There are three main stages of development - egg, nymph and adult. The nymph or hopper stage can be further divided into growth stages called instars, with a moult between each. The following diagram shows the life cycle of the [Australian plague locust](#) which has five instar stages. The times given for development are approximate for optimum conditions during summer.



Locust eggs are laid in the soil. The female drills a hole into the ground using the ovipositor at the tip of the abdomen and lays a 'pod' of eggs which is sealed with froth. The froth helps to protect the eggs from desiccation, disease and predation.



Exposed locust egg pod with white froth plug at soil surface.
The curved shape is typical of shallow autumn laid egg pods

After completing each instar the locust nymph sheds, or moults its skin to continue to grow. On hatching the nymph is wingless but with each successive moult the developing wing buds increase in size and these can be used to determine the [growth stages of nymphs](#).

The final moult into the adult stage is known as fledging, when the locust develops fully formed flying wings. It takes a few weeks before the young adults of most species lay eggs.

Green vegetation is necessary for nymph and adult survival, adult migration and egg development.

How long it takes for a locust to reach maturity depends on the species, habitat conditions and temperature. Nymphs and adults are able regulate their body temperature by basking in the sun or moving to shade.

Further details are available on the life cycle of the economically important species, the [Australian plague locust](#), [spur-throated locust](#) and [migratory locust](#).

Natural enemies of locusts

Egg parasites

Several species of small wasps (3-5 mm long) belonging to the genus *Scelio* parasitise the eggs of locusts and grasshoppers. *Scelio fulgidus* is the most common species in inland areas and is the principal egg parasite of the Australian plague locust. Populations of *S. fulgidus* increase during locust outbreaks and can cause very high egg mortality.

The female burrows through the froth plug on top of the locust egg pod and uses a retractable, needle-like ovipositor to lay an egg in each locust egg. On hatching, the wasp larva feeds inside the egg and kills the locust embryo. In dry conditions *S. fulgidus* adults can remain dormant in the locust egg for a month or more before emerging after rain.



Adult female *Scelio fulgidus*

Flies (Diptera)

Various species of fly are parasites of locusts. They are usually seen in higher rainfall areas and can kill or reduce the fertility of locusts.

Blaesoxipha spp. (sarcophid blowflies) are a common parasite with a rapid larval development of 4-6 days in summer. The female fly deposits minute maggot-like larvae directly on the locust which bore through the cuticle. The fully grown larva emerges through the locust body wall and pupates in the soil. The parasite has several generations a year. It usually does not kill the adult, but parasitised females lay fewer eggs. Nymphs generally die following emergence of the parasite.

Ceracia fergusoni (tachinid fly) is a parasite that resembles *Blaesoxipha*. The female fly glues eggs directly onto the body of locusts. On hatching a small larva burrows through the base of the egg and enters the host. When fully-fed (10-14 days after hatching) the larva emerges from the locust by pushing through the segments in the body. Pupation takes place in the soil, the adult fly emerging 12-15 days later.

Trichopsidea oestracea (bee-fly) females scatter thousands of eggs on the ground or on standing dead timber such as tree stumps or fence posts. The resulting larvae move around freely and enter a locust or

grasshopper through the spiracle (breathing hole). Generally only one larva develops inside each host and after emerging from the host it burrows into the ground to pupate. Those which pupate in autumn become adults in spring during the locust nymphal development period. There is a smaller second generation in the summer. High levels of parasitism have been recorded in dry seasons. Parasitised locusts do not mature eggs and usually die when the fly larva emerges.

Mites

Tarsonemid mites (tracheal mites) are frequently overlooked as they live on the inside of locust tracheae (breathing organs). Their eggs can be seen on the dorsal surface of the locust when its wings are expanded.

Leptus sp. (body mites) is a red mite larvae that is frequently found attached to the wings and bodies particularly of old locusts. The host is generally not affected.

Nematodes

Nematodes (round worms) are important parasites of locusts and grasshoppers in high rainfall areas, but they are rarely found in drier regions.

Amphimermis sp. is an important parasite of wingless grasshoppers (*Phaulacridium vittatum*). The eggs are laid in the soil and hatching is stimulated by rain. On hatching the worm larvae wriggle to the soil surface where they locate and penetrate a grasshopper or locust. They remain in the host for several weeks, retarding development and hindering sexual maturation. The host is killed when the nematode emerges to return to the soil.



Nematode worm from a wingless grasshopper
(Photo by M.Hill from Baker 1991, NSW Agriculture)

Mermis spp. nematodes lay their eggs on vegetation which may be consumed by locusts. The eggs hatch in the gut and the nematode larva penetrates through the gut wall and feeds in the body cavity, eventually killing the locust on emergence.

Protozoans

Little is known about the protozoan parasites of locusts and they are frequently overlooked. Protozoans live as cysts attached to the lining of the mid-gut of locusts. The gut of a heavily parasitised host is yellow; a large number of parasites will interfere with digestion, reduce female fecundity or even kill the locust.

Fungi, bacteria and viruses

Locusts infected by fungi are rarely seen in the field and this may be due to the general aridity of the environment where outbreaks occur. However, a strain of the fungus *Metarhizium acridum*, originally isolated from spur-throated locusts, was developed as a biopesticide by CSIRO and has been used as a control agent by the APLC since 2001.

Bacteria and viruses that affect Australian locusts are poorly known, but mass mortality in NSW during spring 2010 has been linked to a bacterial infection in dense, moist vegetation during a period of high rainfall.

Predators

Numerous birds, mammals and insects eat locusts but generally they are non-specific feeders and have not been shown to have a large impact on locust numbers during an outbreak. Vertebrate predators, unlike some parasites, are unable to increase rapidly in numbers to take advantage of a locust outbreak and their appetite is quickly satisfied, but some migratory and nomadic birds congregate to feed on high density locusts.

Further reading

Baker, GL, 1991, Parasites of locusts and grasshoppers, Agfact AE2. Department of Agriculture NSW, Rydalmere.

Dangerfield, PC, Austin, AD and Baker, GL. 2001, Biology, ecology and systematics of Australian Scelio: wasp parasitoids of locust and grasshopper eggs, CSIRO Publishing, Collingwood, Victoria.

Source: https://www.agriculture.gov.au/biosecurity-trade/pests-diseases-weeds/locusts/about/about_locusts